EWB CLOUD LABS

AWS VPC PEERING

Student Project Report

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Project	Private connectivity between two Amazon VPCs
AWS Region	Asia Pacific (Sydney) — ap-southeast-2
Completion date	29 October 2026
Primary services	Amazon VPC, VPC Peering, EC2, route tables, security groups, network ACLs, Internet Gateway

Project objective

Create a private network connection between two non-overlapping VPCs and verify that two private EC2 workloads can communicate using private IPv4 addresses. Use a public bastion host in VPC A for administrative SSH access.

This report records the actual lab configuration and test results described during the build. The App-B subnet differs from the workbook example; the deployed values are documented transparently below.

1. Project overview

The lab demonstrates VPC peering as a private, one-to-one connection between two VPCs in the same AWS account and Region. Peering does not automatically create routes: each VPC's route tables must contain routes for the other VPC's CIDR, and security controls must permit the intended traffic.

Logical architecture

Administrator laptop	SSH through Bastion (TCP/22)
Bastion — public subnet in VPC A	Public IPv4: 3.107.177.160 Private IPv4: 10.10.1.209
VPC A — 10.10.0.0/16	App-A private subnet: 10.10.2.0/24 App-A private IP: 10.10.2.10
VPC Peering — Active	pcx-0ff23feec843a2429
VPC B — 10.20.0.0/16	App-B private subnet: 10.20.2.0/24 App-B private IP: 10.20.2.10

Architecture note: App-A and App-B were accessed over private IPs. Neither private instance had a public IPv4 address in the configuration recorded during the lab.

2. Resource inventory

Resource	Identifier / value	Notes
VPC A	ewb-vpc-a<br/>vpc-0c672984ed110d8b2	CIDR 10.10.0.0/16
VPC B	ewb-vpc-b<br/>vpc-0c7ab19ace7fc8db3	CIDR 10.20.0.0/16
Peering	pcx-0ff23feec843a2429	ewb-peer-connection; Active
Bastion	3.107.177.160 / 10.10.1.209	Public subnet in VPC A
App-A	10.10.2.10	Private instance in VPC A
App-B	10.20.2.10	Private instance in VPC B

Actual deployment vs workbook

The workbook example uses App-B subnet 10.20.1.0/24 and example host IP 10.20.1.10. In this deployed lab, App-B is in subnet 10.20.2.0/24 with private IP 10.20.2.10. This report uses the actual deployed subnet and host IPs rather than presenting the example values as if they were deployed.

3. Subnets and route tables

Subnet plan

VPC	Subnet name / role	Subnet CIDR	Subnet ID / route table
VPC A	ewb-public-subnet-1 (Bastion)	10.10.1.0/24	subnet-040b447146f6f509b<br/>rtb-07e2002cf4da77cac
VPC A	ewb-private-subnet-1 (App-A)	10.10.2.0/24	subnet-00c11c79142d937e2<br/>rtb-0cd6c2887c65c0203
VPC B	ewb-private-subnet-1 (App-B)	10.20.2.0/24	subnet-03c48fc89472d55de<br/>rtb-05c5e0312358ec8d0
VPC B	ewb-public-subnet-1 (name in console)	10.20.1.0/24	subnet-0405abcae8dc97d87<br/>associated with VPC B m

The VPC B subnet named “ewb-public-subnet-1” is documented as named in the console. Its name alone does not mean it has public internet routing; it was not the App-B subnet used for this test.

Route table entries verified/configured

Route table	Destination	Target	Status / purpose
VPC A public<br/>rtb-07e2002cf4da77cac	10.10.0.0/16	local	Active; local VPC traffic
VPC A public<br/>rtb-07e2002cf4da77cac	0.0.0.0/0	igw-076381f570cd01c27	Internet route for Bastion
VPC A public<br/>rtb-07e2002cf4da77cac	10.20.0.0/16	pcx-0ff23feec843a2429	Peering path for Bastion to VPC B
App-A associated RT<br/>rtb-0cd6c2887c65c0203	10.10.0.0/16	local	Existing local route
App-A associated RT<br/>rtb-0cd6c2887c65c0203	10.20.0.0/16	pcx-0ff23feec843a2429	Added during troubleshooting; Active
VPC B private<br/>rtb-05c5e0312358ec8d0	10.20.0.0/16	local	Active; local VPC traffic
VPC B private<br/>rtb-05c5e0312358ec8d0	10.10.0.0/16	pcx-0ff23feec843a2429	Active; return path to VPC A

Troubleshooting finding

The route table actually associated with App-A's subnet initially contained only the local route. The missing destination 10.20.0.0/16 was added with target pcx-0ff23feec843a2429. After route and security-rule checks, cross-VPC ping succeeded.

4. Security and access

Security controls

Control	Observed / configured details
Private workload addressing	App-A 10.10.2.10 and App-B 10.20.2.10 were accessed using private IPv4 addresses; neithe
Bastion SSH	SSH from the administrator laptop to Bastion public IP 3.107.177.160, then SSH to private Ap
App-A subnet NACL	Inbound and outbound rules showed rule 100: All traffic, 0.0.0.0/0, Allow; default * rule: Deny.
Security groups	ICMP and SSH rules were reviewed/adjusted during troubleshooting. The exact final rule list

5. Validation results

Test	Command / method	Result
Bastion → App-A SSH	SSH from laptop via Bastion to 10.10.2.10	PASS — private shell opened
Bastion → App-B SSH	SSH from laptop via Bastion to 10.20.2.10	PASS — private shell opened
App-B → App-A ping	ping -c 4 10.10.2.10	PASS — 4 transmitted, 4 received, 0% loss
App-A → App-B ping	ping -c 4 10.20.2.10	PASS — 4 transmitted, 4 received, 0% loss
Controlled ICMP failure/recovery	Temporarily block the relevant ICMP rule, test failure, restore the rule, retest	Use here, report completed successfully; insert evidence

Sample successful output summary

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App-B → App-A: 4 packets transmitted, 4 received, 0% packet loss; average RTT about 0.178 ms.
App-A → App-B: 4 packets transmitted, 4 received, 0% packet loss; average RTT about 0.215 ms.
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Why both routes matter

When App-A sends traffic to 10.20.2.10, its subnet route table sends 10.20.0.0/16 traffic through the peering connection. The reply from App-B needs the reciprocal route for 10.10.0.0/16 in VPC B. Peering status alone does not guarantee connectivity; routes, security groups, network ACLs, and host addresses must all be correct.

6. Evidence / screenshot register

Screenshots were reported as captured by the student. Insert them into the final version of this report or attach them as instructed by the instructor. Label each image with the host, test and result. Do not include private keys, passwords, access tokens or other credentials.

No.	Screenshot to include	Caption / result
1	VPC A and VPC B overview with CIDRs	Non-overlapping VPCs: 10.10.0.0/16 and 10.20.0.0/16
2	VPC peering connection details	pcx-0ff23feec843a2429 — Active
3	All three route tables and subnet associations	Forward and return routes configured
4	App-A and App-B EC2 details	Private IPs shown; no public IPv4 on private workloads
5	Bastion SSH / App-A shell / App-B shell	Private SSH access through Bastion
6	App-A → App-B ping terminal	4 received; 0% packet loss
7	App-B → App-A ping terminal	4 received; 0% packet loss
8	Controlled ICMP failure	Ping fails while ICMP rule is temporarily removed/blocked
9	Recovery after restoring ICMP	Ping succeeds again after rule restoration
10	Cleanup evidence	Lab resources removed; no unexpected lab resources remain

7. Conclusion and learning

The lab established private connectivity between two non-overlapping VPCs using VPC peering. The key troubleshooting lesson was to verify the route table actually associated with the workload subnet: App-A's associated route table was missing the route to VPC B. Adding the peering route, checking the reciprocal route and allowing the required traffic enabled successful two-way ping. Private SSH access through a Bastion was also validated.

Cleanup status

Complete this section after cleanup. Confirm termination of the three lab EC2 instances, removal of lab-only EBS volumes/snapshots and any allocated Elastic IP, deletion of the peering connection, lab security groups, subnets, custom route tables and Internet Gateway, and deletion of the two lab VPCs. Check the Sydney Region and Billing/Cost Explorer for any remaining chargeable resources. Do not delete unrelated/default VPC resources.

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